

<i>index, n, i, j, k</i>	index metavariables
<i>int, v</i>	integer literal
<i>ident, x</i>	variable identifiers
<i>fident, f</i>	function identifier
<i>tvar, T</i>	type-level variable identifiers

lin	$::=$ $ $ n $ $ T $ $ $lin + lin'$ $ $ $lin - lin'$ $ $ $n * T$	Linear arithmetic expression integer literal integer variable linear addition linear subtraction linear product
pow, p	$::=$ $ $ 1 $ $ 2 $ $ $2 \wedge lin$	Power of two one two linear exponent of two
$size, s$	$::=$ $ $ n $ $ p $ $ s/p $ $ $s + s'$ $ $ $s - s'$ $ $ $s * s'$	Sizes as dyadic numbers constant size power of two size division of size power of two addition of sizes subtraction of sizes multiplication of sizes
$range, R$	$::=$ $ $ $s..s'$ $ $ $s, p..s'$	range of numeric indices Range with unit step Range with step power of two
$tkind, K$	$::=$ $ $ Group $ $ Field $ $ Size	Type kinds Elliptic curve group Finite field modulo prime Size type variable
$tbase, B$	$::=$ $ $ Scalar T $ $ Pairing $T_1 T_2$ $ $ Additive T $ $ Multiplicative T $ $ T	Base types Scalar field of group Pairing-friendly groups Additive subgroup of field Multiplicative subgroup of field Type variable
$type, t$	$::=$ $ $ Uni $\langle B, s \rangle$ $ $ Mle $\langle B, s \rangle$ $ $ $[t; s]$ $ $ Fin $\langle R \rangle$ $ $ B	Static types Univariate polynomial Multilinear polynomial Vector type Index type Base type
$tdecl, TD$	$::=$ $ $ $T : K$	Type variable of given kind
$tdecls, TDs$	$::=$ $ $ $TD_0, \dots, TD_n$	Type variable declarations

<i>qualifier, q</i>	$::=$ public private	Visibility qualifier Publicly visible argument Private argument
<i>arg</i>	$::=$ $q\ x : t$	Argument Typed, qualified argument
<i>args</i>	$::=$ $arg_0, \dots, arg_n$	Arguments Argument list
<i>bin_op, op</i>	$::=$ $+$ $-$ $*$ $\cdot$ $/$ $\wedge$ $++$	Zippel binary operators Addition Subtraction Product Inner product Division Exponentiation Vector concatenation
<i>exp, e, d</i>	$::=$ v x $e\ op\ d$ gen $\langle T \rangle$ coef (e) mle (e) interpolate (e, d) random $\langle t \rangle$ challenge $\langle t \rangle$ $[e_0, \dots, e_n]$ $[e\ \text{for}\ x\ \text{in}\ r]$ $x[s]$ $f(e_0, \dots, e_n)$ reduce $(op\ \text{"}, "$ aexp $)$ assert $(b); e$ verify $(b); e$ let $x := e\ \text{in}\ d$ $x \leftarrow e; d$	Zippel expressions Field literal Variable Binary operation Group generator Univariate polynomial coefficients Multilinear evaluations over bool b Lagrange interpolation from points Sample pseudorandom number gen Generate a verifier challenge Vector of expressions Map comprehension index a vector Function application Reduce binary operation over vect Prover assertion Verifier assertion Let binding Transcript binding
<i>bexp, b</i>	$::=$ $e = e'$ $f(e_0, \dots, e_n)$ $b \ \&\&\ b'$ $b \ \ b'$	Boolean expressions Equality of expressions Protocol application Boolean and operation Boolean or operation
<i>decl, D</i>	$::=$ proto $f \langle TDs \rangle (args)\ \text{where}\ b := e; \text{verify}\ (b').$ $f_n\ f \langle TDs \rangle (args) \rightarrow t := e.$	Top-level declarations Zippel protocol Zippel function